

SV & UVM Practice Questions

Part 6: Technical Interview Preparation

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1. What is a "class"?
2. Explain about various constraints in SV.
3. What is an assertion?
4. Explain about various data types of SV.
5. What is a dynamic Array?
6. Explain about clocking block and program block.
7. What is polymorphism?
8. What is the difference between code coverage and functional coverage?
9. Consider, there are 3 classes namely class A, class B and class C. Class B is extended from class A and class C is extended from class B. There is a print method for all classes, how class A can call the print method from all classes?
10. Explain about the conditional operators in SV.
11. Explain about various loop operators in SV.
12. Explain the difference between 2 state and 4 state datatypes.
13. What is the significance of super.new?
14. Explain about the OOP concepts.
15. How will you find glitch in verification?
16. Explain the difference between a program block and a module.
17. How can a function return more than one value?
18. Write the syntax of dynamic array and associative array.
19. What is an Interface?
20. What is the race condition? How can it be resolved?
21. What is a Mod Port?
22. What are assertions? Write an assertion for T-FF and write the constraint for the same.
23. What is a virtual interface? How to connect it?
24. Explain the difference between shallow copy and deep copy.
25. Explain this and super keywords.
26. What is the need for constraints?

27. What are the different types of constraints?
28. Explain distribution constraint.
29. What is a soft constraint?
30. What are the different types of coverage?
31. What is the difference between illegal and ignore bins?
32. What is the difference between associative array and dynamic array?
33. Explain the difference between pass by value and pass by reference.
34. What is the difference between get and peek methods of mailbox?
35. What is the difference between synchronous and asynchronous RAM? What is the application of dual port RAM? Can it be used for pipelined operations? What is the disadvantage of single port ram?
36. Write an assertion to check even number of 1's in a 7-bit data.
37. Write an assertion to verify clock period.
38. How to sign-off verification?
39. What is functional coverage and why do we need it?
40. What is a covergroup and a coverpoint?
41. Why do we need randomization?
42. Explain fork join, fork join_any and fork join_none with examples.
43. What is Inheritance and polymorphism?
44. What is a Mailbox?
45. Why do we prefer mailbox over queues?
46. What are the default values of int, byte, bit?
47. How will you randomize a number with an even number of one's?
48. What is the difference between bit and logic?
49. What is the difference between local, private, protected and public?
50. Write a checker to identify a glitch where at every 10 ns glitch appears.
51. What is Virtual interface and how it is used?
52. What is the difference between code coverage and functional coverage?

53. What is the difference between queue and mailbox?
54. What is the difference between dynamic array and associative array?
55. What is the difference between logic and reg datatype?
56. What is a Lint?
57. Explain the methods of queue and mailbox.
58. What is the difference between pass-by-value and pass-by-ref?
59. What is the difference between dynamic memory allocation and static memory allocation?
60. How can you sample a cover group without using sample () method?
61. What is \$write in SVA?
62. Interface is static, how do you declare it in class?
63. What are the advantages of SV over Verilog?
64. Explain the difference between task and function in System Verilog.
65. What are the different types of assertions?
66. What are the different system tasks used in SVA?
67. How do you turn off randomization?
68. Write an SV constraint to generate even numbers in an array.
69. How can a task be called inside a function if necessary?
70. Explain about the different regions in SV event scheduler?
71. How do you connect a driver and a generator in SV?
72. What are the methods used in mailbox?
73. What can we use in place of mailbox?
74. Write a constraint to generate a random array and generates same value then after again and again.
75. What is Constrained Random Coverage Driven Verification?
76. Write a design to verify all the 20 consecutive numbers are unique, assume a 10 bit number.
77. Write a TB to verify all the 20 consecutive numbers are unique, assume a 10 bit number.
78. Write a design which generates the odd bit 2's complement out for any given input.

79. A design takes exactly the input no of bit's clock cycles to produce output and no input signals are produced within this clock cycles as it is a violation to the design. Design does a right rotate shift and produce the output. Write a code for this design.
80. Write a test bench to verify the design and write the scoreboard for the design.
81. Write an example for deep copy and shallow copy.
82. Write a design to generate the CRC for any bit's input given the polynomial being $X^4 + X^2 + 1$ pad the CRC to the input and generate the output.
83. Write a design to produce a sorted output for every 10 value input.
84. Write a design to generate an output of 1 if the no of ones in odd value of the input are even and 0 if no so.
85. **Scenario:** There is a class consisting of four class members (real value, enum value, int value, and string value). While randomizing the class, the user should be able to randomize it according to their needs. No other class should be able to access the enum value and no other class other than the derived class should be able to access the String value. Write a code based on the above requirement.
86. Write a task to convert a binary value to a decimal value and vice-versa (also for hexa and octa).
87. Write a class to behave like a mail box.
88. Write a FILO design for two value with minimum code.
89. Implement a functionality using rand that should work like randc.
90. Let there be a phase mechanism in the SV and assume there is a parent class such that if you call the parent class handle and call any phase it should jump to that phase.
91. Write a task to identify the frequency of any input clock for the task.
92. Write a SRAM design with data being a bi-directional and a test bench for the same.
93. Find the output of f.show and s.show and also find output without virtual keyword.

94. A signal "req" is asserted after 5 cycles of "ack", req is single clock pulse and ack is single clock pulse. Write the checker in Verilog.

95. Find the output of the following snippet:

```
`define MACRO(x) x << 1;
initial
begin
$display(">>> ", `MACRO(1) + 5);
end
```

96. Write a program x! divisible by y using the func(unsigned int x,y)?

97. What is the difference between \$rose(a) and \$posedge(a)?

98. What is the difference between sub-routine and built-in functions?

99. Write a SVA to check that if signal A is HIGH, then after one clock cycle signal B is asserted only for one clock cycle.

100. What is the difference between fork join and fork...join_none?

101. Why fork...join_none is used in SV TB architecture? Can we use fork...join_any?

102. Write an assertion to find a glitch.

103. What is the difference between unidirectional and bidirectional Driver?

104. 3 processes are there inside fork-join, delays are random inside processes, Write logic to detect which process is faster.

105. You have 2 drivers namely Driver A and Driver B, if you want your Driver A to interact first and then the Driver B how the synchronization is achieved?

106. If we have 2 drivers in our testbench architecture where will the semaphore be declared?

107. Write a sv function to realize the logic shown below. Input is dynamic single dimensional array.

108. What is class?

109. What is Inheritance?

110. What is the difference between rand and randc?

111. What are semaphores?
112. What is a mailbox?
113. What is randomization? What are the use constraints? Explain with some examples.
114. What are clocking blocks and modports?
115. Write a code and explain cross coverage.
116. How do we connect multiple interfaces to a single DUT?
117. What is code coverage?
118. What is a coverpoint?
119. What is coverage driven verification?
120. What components are there in a SystemVerilog testbench?
121. What do we do in a scoreboard?
122. Explain about various mailbox methods.
123. Explain about queues and associative arrays.
124. Which is better? Pass by value or Pass by reference?
125. Explain about polymorphism and interface.
126. What is an Assertion?
127. If code coverage is of 90% and functional coverage is of 100%. What does this indicate?
128. Explain the difference between fork-join, fork-join_any, and fork-join_none.
129. When there are two parallel blocks, how can you ensure that one block is executed after the other block?
130. What is the meaning of `@(posedge clk) a | => b;`
131. Write a constraint to randomize 3 variables a, b, and c that a will be randomized independently, b will be randomized only after a, and c will be randomized after b.
132. What is objection in SV?
133. Explain about stratified event queue.
134. What is the output of following snippet? `(q.size=10; for loop with pop_back)`.
135. Why are we writing the compare logic in scoreboard not in interface.
136. How do you generate parity packet for router in test bench?
137. Write a program to display Fibonacci series.

138. Write a program to reverse the number from an array.
139. What is the difference between `|->` and `##0` in assertion?
140. Why do we use clocking blocks?
141. Write an assertion to detect the following condition. The signal `req` should be enabled within 10 clock signals.
142. Write an assertion for glitch detection.
143. Can TLM ports be used in SV? If so, how?
144. What is the difference between static and dynamic memories?
145. Write a program to find the missing number `int array[ ] = {1,3,2,5,9,4,7,6}`
146. What are the various simulation software's used for verification in industry?
147. What are the other types of verifications in the industry?
148. How will we know verification is complete? What are the checklists?
149. What are the components required to be developed in verification projects?
150. What is set-up time and hold time?
151. What is timescale directive?
152. What are the features we are going to verify if we develop any testbench?
153. How to kill a thread or disable a thread?
154. What is meant by coverage driven verification?
155. What is the difference between `$random` and `$urandom`?
156. What is an Abstract class?
157. Explain the different types of fork join threads and how will you come out of a fork join thread?
158. What is static function and automatic function?
159. Why TLM port is needed in spite of having mailbox?
160. What is Factory overriding? Why it is not used in system Verilog?
161. Why do we use 'create' method in UVM rather than using 'new' constructor?
162. Explain about `config_db`.
163. Explain the difference between object and component in UVM.
164. What is the significance of virtual sequence?

165. Why is sequence an object? Why cannot we generate stimulus using component?
166. What is a stimulus? Both configuration and sequence are objects then why cannot we write configuration variable in sequence?

THANK YOU